

Stored Procedures in Amazon Redshift

To Begin with the Lab

Summary of the Lab

In this lab, a stored procedure named `update_sales_price` was created using PL/pgSQL in **Amazon Redshift** to automate updates on the sales table. The procedure accepted a sales ID and a new price value as parameters, then updated the corresponding `pricepaid` value after validating whether the sales record existed. Both invalid and valid sales IDs were tested to verify the validation logic and update functionality. The stored procedure was successfully created and became visible under the Stored Procedures section in the Query Editor. Finally, the updated sales record was validated using SQL queries to confirm that the new value had been applied successfully.

- Sign in to the AWS Management Console
- Navigate to **Amazon Redshift** → open **Redshift Query Editor v2**
- Connect to the sample Redshift cluster
- Select the dev database and public schema

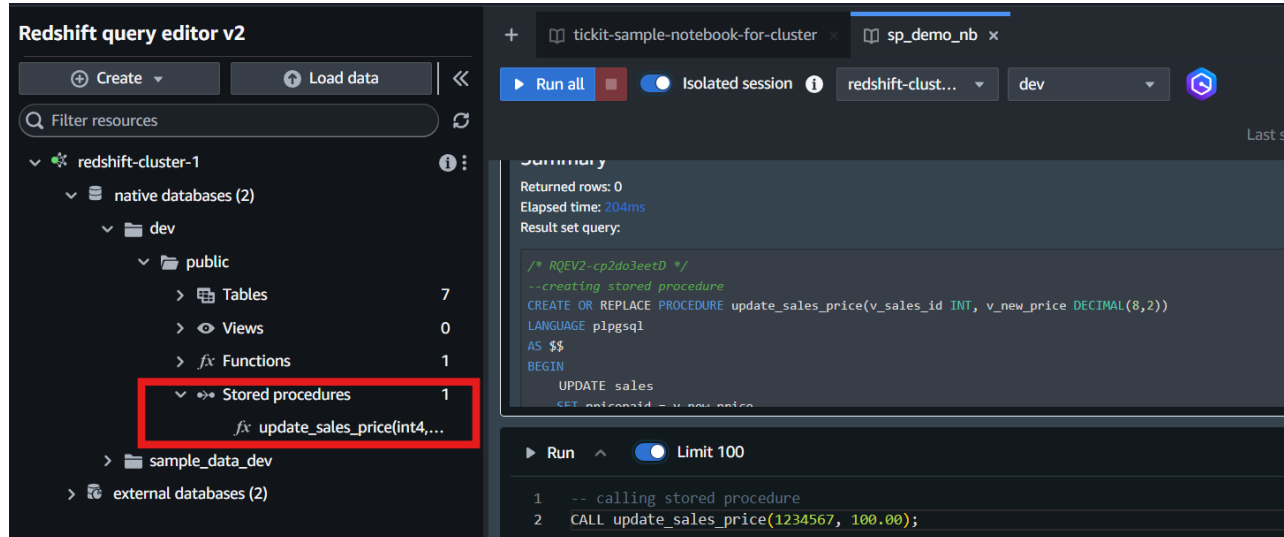
```
CREATE OR REPLACE PROCEDURE update_sales_price(v_sales_id INT, v_new_price DECIMAL(8,2))
LANGUAGE plpgsql
AS $$
BEGIN
```

Field	Type	NL	CMP
salesid	integer	NN	az64
listid	integer	NN	az64
sellerid	integer	NN	az64
buyerid	integer	NN	az64
eventid	integer	NN	az64

Summary
Returned rows: 0
Query ID: 5319
Elapsed time: 304ms
Result set query:

- Open the sales table from dev → public → sales

- Review important columns:
 - salesid
 - pricepaid
- Create the stored procedure

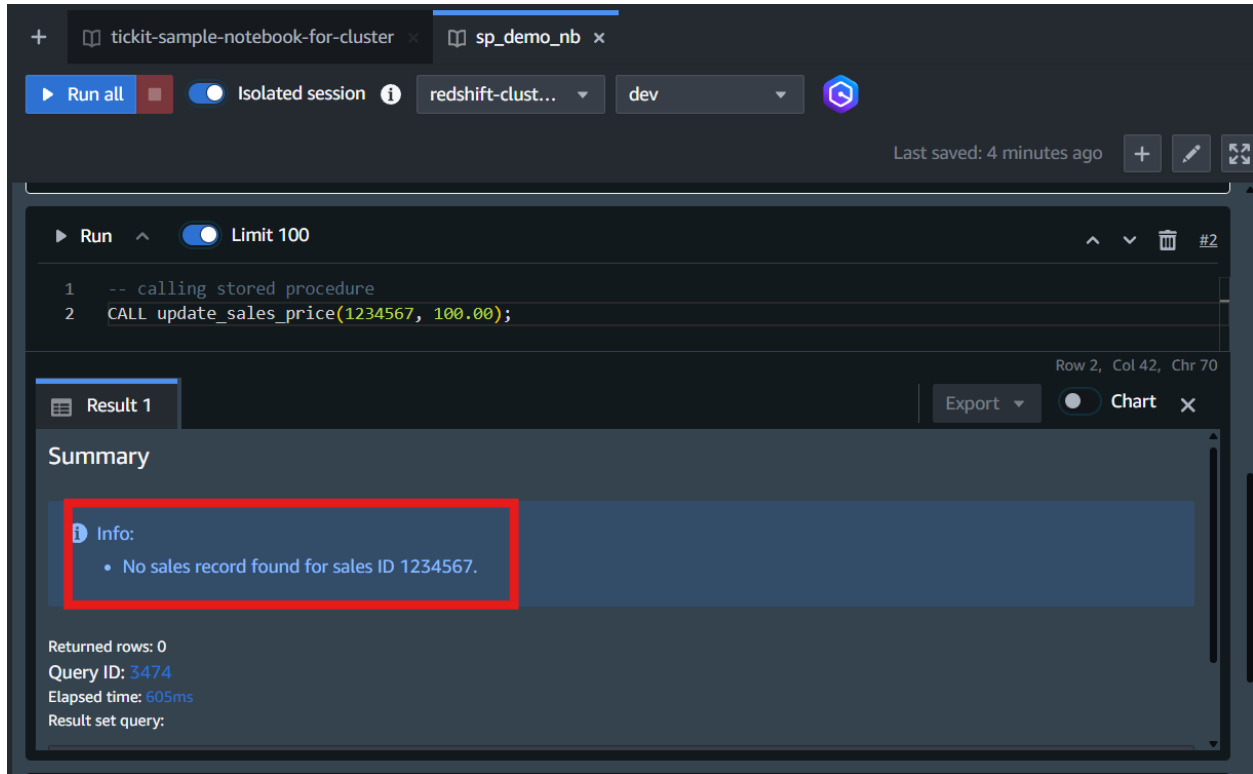


```
CREATE OR REPLACE PROCEDURE update_sales_price(
  v_sales_id INT,
  v_new_price DECIMAL(8,2)
)
AS $$
BEGIN
  UPDATE sales
  SET pricepaid = v_new_price
  WHERE salesid = v_sales_id;

  IF NOT FOUND THEN
    RAISE NOTICE 'No sales record found for provided sales ID';
  END IF;
END;
$$ LANGUAGE plpgsql;
```

- Click **Run all** to execute the procedure creation script
- Refresh the left navigation panel
- Verify that the Stored Procedures section now contains update_sales_price

- Execute the procedure using an invalid sales ID
CALL update_sales_price(1234567, 100.00);
- Verify that the output shows the validation notice:



No sales record found for provided sales ID

- Execute the procedure using a valid sales ID

tickit-sample-notebook-for-cluster x sp_demo_nb x

Run all Isolated session redshift-clust... dev

Last saved: 4 minutes ago

Elapsed time: 605ms
Result set query:

```
/* RQEV2-rnCbfv17VR */  
-- calling stored procedure  
CALL update_sales_price(1234567, 100.00)
```

Run Limit 100 #3

```
1 -- select * from sales where salesid=147292;  
2 select * from sales where salesid=123456;
```

Result 1 (1) Export Chart X

salesid	listid	sellerid
123456	141323	5708

Run status

1 out of 1 cell succeeded

Start time: 8:47:11 pm (UTC+05:30)

✓ #3 275 ms

CALL update_sales_price(123456, 3310.00);

- Verify that the procedure executes successfully
- Run a validation query on the updated sales record

SELECT *

FROM sales

WHERE salesid = 123456;

Run Limit 100 #3

```
1 -- select * from sales where salesid=147292;  
2 select * from sales where salesid=123456;
```

Result 1 (1) Export Chart X

buyerid	eventid	dateid	qtysold	pricepaid
15130	2497	2101	2	3310

Row 2, Col 42, Chr 87

Query ID: 3669 Elapsed time: 32 ms Total rows: 1

- Verify that the pricepaid value is updated to 3310
- Confirm that the stored procedure updated the sales record successfully